

Stochastic Hybrid Dynamics Across Symmetry Strata

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Abstract

This talk presents a two-part development of stochastic hybrid systems whose continuous state spaces may have different dimensions.

In the first part, we consider stochastic hybrid systems in which a continuous diffusion evolves within each mode until it reaches a guard, where a reset may transfer the state to a mode of lower dimension. A measure-theoretic formulation describes this transfer by pushing the outgoing probability flux through the reset map. This yields coupled Fokker-Planck equations that conserve total probability across modes of different dimensions. We also introduce the corresponding stochastic Koopman and Frobenius-Perron operators on manifolds, providing complementary observable- and density-based descriptions of the hybrid dynamics.

In the second part, we use this framework to construct generative dynamics across symmetry strata. A forward process moves toward lower-dimensional strata satisfying additional symmetry constraints. Its exact time reversal produces stochastic births of degrees of freedom and thereby allows individual trajectories to break symmetry while preserving equivariance in law. The forward boundary flux jointly determines the birth time, branch selection, and entrance into the higher-dimensional stratum. Under suitable geometric and regularity assumptions, the reversed process is well defined, nonexplosive, and recovers the complete forward path law in reverse. We conclude with learning principles and numerical studies ranging from an analytic triangle benchmark to wallpaper and crystallographic symmetry families.

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